Technical Comparison of BM25 and Dense Vector Search for Document Retrieval

Overview

BM25 is a term-frequency-based probabilistic information retrieval function that relies on exact token matching, inverse document frequency, and term saturation. Dense vector search uses neural embeddings generated by pre-trained language models to map documents and queries into a continuous vector space, retrieving items based on semantic similarity via distance metrics such as inner product or cosine similarity. While BM25 excels at precision keyword matching and rare-term lookup, dense vector search excels at capturing conceptual overlap, paraphrasing, and semantic intent.

Comparison

Criterion	Approach A	Approach B
Query-Document Matching Mechanism	BM25 relies on exact lexical token matching and frequency statistics (TF-IDF variants) within documents.	Dense vector search relies on continuous vector representations, matching queries and documents by semantic proximity in high-dimensional space.
Computational Efficiency & Indexing	BM25 is lightweight, highly optimized in inverted indexes (e.g., Lucene), and requires minimal memory and CPU resources.	Dense vector search requires neural network inference for embedding generation, high-memory vector databases, and approximate nearest neighbor (ANN) graph indices.
Handling OOV and Synonyms	BM25 fails to match synonyms or out-of-vocabulary terms unless explicit query expansion or synonym dictionaries are applied.	Dense vector search naturally handles synonyms, paraphrases, and multi-lingual queries due to dense semantic clustering.

Strengths & Weaknesses

Approach A

+ Extremely fast indexing and low query latency on standard CPU hardware
+ High precision for exact keyword, part number, entity name, and code snippet lookups
+ Interpretable term weights and transparent scoring mechanics
+ No dependency on heavy GPU hardware or pre-trained transformer inference models
- Inability to understand semantic context or paraphrased variations of search intent
- Fails completely when query terms do not lexically overlap with document terms (vocabulary mismatch problem)
- Sensitive to spelling errors and variations in terminology

Approach B

+ Captures deep semantic meaning, contextual nuances, and conceptual questions
+ Resolves the vocabulary mismatch problem by mapping different terms with similar meanings close together
+ Supports cross-lingual and paraphrase-robust retrieval tasks seamlessly
- High computational cost for generating embeddings and maintaining approximate nearest neighbor (ANN) indices
- Prone to hallucination or false-positive semantic matches on rare, highly specific alphanumeric strings
- Requires significant storage overhead for high-dimensional float vectors

Recommendation

> For enterprise search, question-answering systems, and Retrieval-Augmented Generation (RAG), a hybrid search architecture combining BM25 and dense vector search via reciprocal rank fusion (RRF) is recommended. This hybrid configuration captures both the exact lexical precision of BM25 and the robust semantic recall of dense vector embeddings, mitigating the individual weaknesses of each approach.

Sources

Integrating the Probabilistic Models BM25/BM25F into Lucene
Establishes BM25 as the foundational state-of-the-art term-based information retrieval model integrated into enterprise search engines.
https://arxiv.org/abs/0911.5046v2

Fine-tune the Entire RAG Architecture (including DPR retriever) for Question-Answering

Demonstrates the integration and optimization of dense passage retrieval (DPR) models within neural retrieval-augmented generation pipelines.

https://arxiv.org/abs/2106.11517v1

Vendi-RAG: Adaptively Trading-Off Diversity And Quality Significantly Improves Retrieval Augmented Generation With LLMs

Analyzes how retrieval strategies affect the balance of diversity and quality in downstream LLM generation tasks.

https://arxiv.org/abs/2502.11228v2